# Lecture 15
# Refinement Types

With slides from Nadia Polikarpova

# Non-trivial properties with refinement types

$$\text{replicate} :: n\text{: Int} \rightarrow x\text{: } \beta \rightarrow v\text{: List } \beta$$

$$\text{replicate} :: n\text{: Nat} \rightarrow x\text{: } \beta \rightarrow \{\, v\text{: List } \beta \mid \text{len } v = n \,\}$$

# Example: insert

Goal type:

$$x: \beta \rightarrow xs: \text{IncList}\ \beta \rightarrow \{\ v: \text{IncList}\ \beta \mid \mathit{elems}\ v = \mathit{elems}\ xs + [x]\ \}$$

Components:

```
data IncList α where
  Nil ::      IncList α
  Cons :: x: α  → xs: IncList { v: α | v ≥ x }
      →       IncList α
```

# Insert solution

Goal type:

x: β → xs: IncList β → { v: IncList β | *elems* v = *elems* xs + [x] }

Solution:

```
insert x xs = match xs with
  Nil → Cons x Nil
  Cons y ys → if x ≤ y
    then Cons x (Cons y ys)
    else Cons y (insert x ys)      :: IncList { v: Int | v ≥ y }
```

# Static Semantics

Typing rules

- Typing rules tell us how to derive typing judgments

$$\frac{premises}{Judgment}$$

Ex. Language of Expressions

$$\frac{x{:}T \in \Gamma}{\Gamma \vdash x : T} \qquad \frac{}{\Gamma \vdash N : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 + e2 : int}$$

# Ex. Language of Expressions

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Show that the following Judgment is valid

$$x{:}int, y{:}int \vdash x + (y + 5) : int$$

$$\frac{x{:}int, y{:}int \vdash x{:}int \quad x{:}int, y{:}int \vdash (y + 5) : int}{x{:}int, y{:}int \vdash x + (y + 5) : int}$$

$$\frac{\dfrac{x{:}int \in x{:}int, y{:}int}{x{:}int, y{:}int \vdash x{:}int} \qquad \dfrac{x{:}int, y{:}int \vdash y{:}int \quad x{:}int, y{:}int \vdash 5 : int}{x{:}int, y{:}int \vdash (y + 5) : int}}{x{:}int, y{:}int \vdash x + (y + 5) : int}$$

# Simply Typed $\lambda$ Calculus ($F_1$)

Basic Typing Rules

$$\frac{x{:}\tau \in \Gamma}{\Gamma \vdash x : \tau} \qquad \frac{\Gamma, x{:}\tau_1 \vdash e{:}\tau_2}{\Gamma \vdash (\lambda x{:}\tau_1\ e){:}\tau_1 \to \tau_2} \qquad \frac{\Gamma \vdash e_1{:}\tau' \to \tau \quad \Gamma \vdash e_2{:}\tau'}{\Gamma \vdash e_1 e_2{:}\tau}$$

Extensions

$$\frac{}{\Gamma \vdash N : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 + e2 : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 = e2 : bool}$$

$$\frac{\Gamma \vdash e{:}bool \qquad \Gamma \vdash e_t : \tau \qquad \Gamma \vdash e_f : \tau}{\Gamma \vdash if\ e\ then\ e_t\ else\ e_f : \tau}$$

# Example

Is this a valid typing judgment?

$$\vdash (\lambda x{:}\, bool\ \lambda y{:}\, int \quad if\ x\ then\ y\ else\ y + 1){:}\, bool \to int \to int$$

How about this one?

$$\vdash (\lambda x{:}\, int \ \ \lambda y{:}\, bool\ x + y){:}\, int \to bool \to int$$

# Example

What's the type of this function?

(λ f. λ x. *if* x = 1 *then* x *else* (f f (x-1) ) * x)

$$\frac{x{:}\tau \in \Gamma}{\Gamma \vdash x : \tau}$$

$$\frac{\Gamma, x{:}\tau_1 \vdash e{:}\tau_2}{\Gamma \vdash (\lambda x{:}\tau_1\ e){:}\tau_1 \to \tau_2}$$

$$\frac{\Gamma \vdash e_1{:}\tau' \to \tau \quad \Gamma \vdash e_2{:}\tau'}{\Gamma \vdash e_1 e_2{:}\tau}$$

$$\frac{}{\Gamma \vdash N : int}$$

$$\frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 + e2 : int}$$

$$\frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 = e2 : bool}$$

$$\frac{\Gamma \vdash e{:}bool \quad \Gamma \vdash e_t : \tau \quad \Gamma \vdash e_f : \tau}{\Gamma \vdash if\ e\ then\ e_t\ else\ e_f : \tau}$$

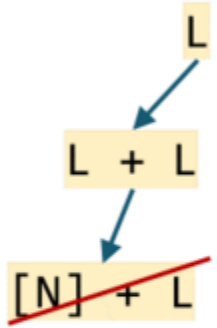
- Hint: This IS a trick question

# Motivation

**Goal:** use deductive reasoning for top-down propagation

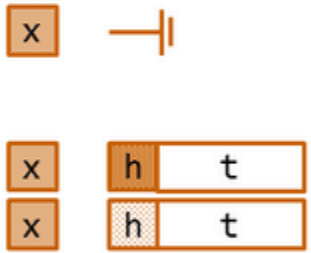
- prune unverifiable candidates early
- need synthesis-friendly verification technique!

**Observation:** type checkers are good at rejecting incomplete programs!

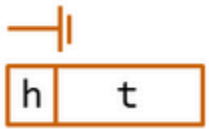


# Running example

```
// Insert x into a sorted list xs
insert :: x:e  →  xs:List e  →  List e
insert x xs =
  match xs with
    Nil →
    Cons h t →
      if x ≤ h
        then Cons x xs
        else Cons h (insert x t)
```




```
data List e where
  Nil :: List e
  Cons :: h:e  →  t:List e  →  List e
```



# Rejecting incomplete programs

[Pierce, Turner. TPLS'00]

```
// Insert x into a sorted list xs
insert :: x:e  →  xs:List e  →  List e
insert x xs =
  match xs with
    Nil → Cons xs ...
    ...
```

Expected
e
and got
List e

bidirectional type-checking!

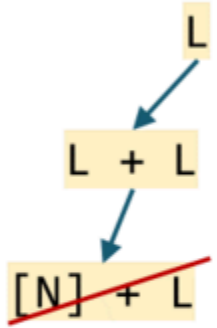
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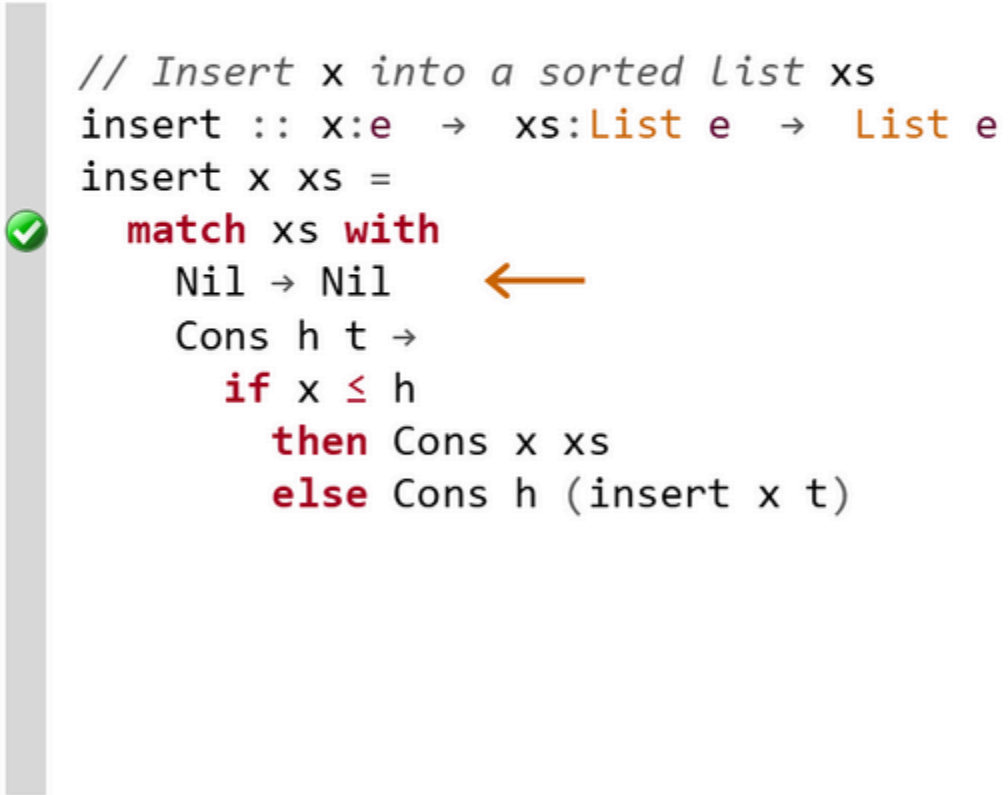
- prune unverifiable candidates early
- need synthesis-friendly verification technique!

**Observation:** type checkers are good at rejecting incomplete programs!

**Idea:** can we use types as behavioral constraints for synthesis?



# Conventional types are not enough



```
// Insert x into a sorted list xs
insert :: x:e  →  xs:List e  →  List e
insert x xs =
  match xs with
    Nil → Nil
    Cons h t →
      if x ≤ h
        then Cons x xs
        else Cons h (insert x t)
```

# Refinement types

[Rondon et al.'08, Kawaguchi et al.'09]

```
Nat
```

base types

```
max :: x: Int → y: Int → { v: Int | x ≤ v ∧ y ≤ v }
```

dependent function types

```
xs :: { v: List Nat }
```

polymorphic datatypes

```
data List α where
    Nil  ::  { List α | len v = 0 }
    Cons  ::  x: α  → xs: List α
                    → { List α | len v = len
                      xs + 1 }
```

```
measure len :: List α → Int
    len Nil = 0
    len (Cons _ xs) = len xs + 1
```

# Refinement types

$$
\begin{array}{lll}
e & ::= \text{true} \mid \text{false} \mid n \mid e + e & \text{Terms} \\
 & \quad \mid x \mid e\ e \mid \lambda x{:}T.\, e & \\
T & ::= \{\nu{:}\ \mathrm{B} \mid \mathrm{e}\} & \text{(basic types)} \quad \text{Types} \\
 & \quad \mid x{:}T_1 \to T_2 & \text{(function types)} \\
 & \quad \mid \alpha & \text{(type variables)} \\
S & ::= T \mid \forall \alpha.\, S & \text{Type schemas}
\end{array}
$$

T-num $$\frac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{\nu{:}\ \mathrm{Int} \mid \nu = n\}}$$

T-abs $$\frac{\Gamma; x{:}T \vdash e :: T'}{\Gamma \vdash \lambda x{:}T.\, e :: T \to T'}$$

T-var $$\frac{(x{:}\{\nu{:}B \mid P(\nu)\} \in \Gamma)}{\Gamma \vdash x :: \{\nu{:}B \mid \nu = x\}} \quad \frac{\Gamma(x) \text{not basic}}{\Gamma \vdash x :: \Gamma(x)}$$

T-app $$\frac{\Gamma \vdash e_1 :: x{:}T \to T' \quad \Gamma \vdash e_2 :: T}{\Gamma \vdash e_1\ e_2 :: T'[x \to e_2]}$$

# Example

$Nat \equiv \{v\colon Int \mid v > 0\}$

$$\text{T-num}\ \frac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{v\colon \text{Int} \mid v = n\}} \qquad \text{T-abs}\ \frac{\Gamma; x\colon T \vdash e :: T'}{\Gamma \vdash \lambda x\colon T.\, e :: T \to T'}$$

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$$incr\colon x\colon Nat \to \{v\colon Int \mid v = x + 1\} \vdash \lambda y\colon Nat.\, incr\ y :: y\colon Nat \to \{v\colon Int \mid v = y + 1\}.$$

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$$\frac{\dfrac{incr{:}\, x{:}Nat \to \{v{:}\,Int \mid v = x+1\};\ y{:}Nat \vdash incr :: x{:}Nat \to \{v{:}\,Int \mid v = x+1\}. \quad incr{:}\, x{:}Nat \to \{v{:}\,Int \mid v = x+1\};\ y{:}Nat \vdash y :: Nat.}{incr{:}\, x{:}Nat \to \{v{:}\,Int \mid v = x+1\};\ y{:}Nat \vdash incr\ y :: \{v{:}\,Int \mid v = y+1\}.}}{incr{:}\, x{:}Nat \to \{v{:}\,Int \mid v = x+1\} \vdash \lambda y{:}Nat.\, incr\ y :: y{:}Nat \to \{v{:}\,Int \mid v = y+1\}.}$$

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$x{:}Nat \to \{v{:}\,Int \mid v = x + 1\}$; is not a basic type

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Basic rules by themselves are not very useful!!

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# Another Example

$Nat \equiv \{v\colon Int \mid v > 0\}$

$$\text{T-num}\ \frac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{v\colon \text{Int} \mid v = n\}} \qquad \text{T-abs}\ \frac{\Gamma; x\colon T \vdash e :: T'}{\Gamma \vdash \lambda x\colon T.\, e :: T \to T'}$$

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$$\text{dbl}\colon x\colon \text{Int} \to \{v\colon \text{Int} \mid v = 2 * x\} \vdash (\lambda y\colon \text{Nat}.\, \text{dbl}\ y)\ 5 :: \text{Nat}$$

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# Subtyping

Intuitively, $T'$ is a subtype of $T$ if all values of type $T'$ also belong to $T$

- written $T' <: T$
- e.g. $\text{Nat} <: \text{Int}$ or $\{v: \text{Int} \mid v = 5\} <: \text{Nat}$

Defined via inference rules:

Sub-base $$\frac{[\![\Gamma]\!] \wedge e' \Rightarrow e}{\Gamma \vdash \{v: B \mid e'\} <: \{v: B \mid e\}}$$

Sub-fun $$\frac{\Gamma \vdash T_1 <: T_1' \qquad \Gamma; x: T_1 \vdash T_2' <: T_2}{\Gamma \vdash x: T'_1 \to T_2' <: x: T_1 \to T_2}$$

$$\frac{\Gamma \vdash e: T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e: T}$$

# Constraints from the environment

We saw the expression $[\![\Gamma]\!]$ in the rule

- This extracts constraints from the environment

$$[\![\Gamma]\!] = \bigwedge\{e \mid e \in \Gamma\} \wedge \bigwedge\{[\![e[v \to x]]\!] \mid x{:}\{v{:}B \mid e\} \in \Gamma\ \}$$

- The environment can store constraints $e$, which get included.
- The environment also contains variables $x{:}\{v{:}B \mid e\}$.
  All their constraints must also be included, with suitable renaming inside the constraint.

# Example

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Sub-base $$\frac{[\![\Gamma]\!] \wedge e' \Rightarrow e}{\Gamma \vdash\ \{v{:}\ B \mid e'\} <:\ \{v{:}\ B \mid e\}}$$

$$\frac{\Gamma \vdash e{:}T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e{:}T}$$

$$incr{:}\, x{:}Nat \to \{v{:}Int \mid v = x + 1\};\ y{:}Nat \vdash y :: \{v{:}Int \mid v = y\}.$$

$$incr{:}\, x{:}Nat \to \{v{:}Int \mid v = x + 1\};\ y{:}Nat \vdash \{v{:}Int \mid v = y\} <: \{v{:}Int \mid v > 0\}$$

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$$Nat \equiv \{v{:}\, Int \mid v > 0\}$$

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$$\text{dbl}{:}\ x{:}\text{Int} \to \{v{:}\text{Int} \mid v = 2 * x\} \vdash (\lambda y{:}\text{Nat}.\,\text{dbl}\ y) :: Nat \to Nat \qquad \text{dbl}{:}\ x{:}\text{Int} \to \{v{:}\text{Int} \mid v = 2 * x\} \vdash 5 :: Nat$$

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$$\dfrac{\dfrac{}{\text{dbl: } x{:}\text{Int} \to \{v{:}\text{Int} \mid v = 2 * x\} \vdash 5 :: \{v{:}Int \mid v = 5\}} \qquad \dfrac{v = 5 \Rightarrow v > 0}{\{v{:}Int \mid v = 5\} <: \{v{:}Int \mid v > 0\}}}{\text{dbl: } x{:}\text{Int} \to \{v{:}\text{Int} \mid v = 2 * x\} \vdash (\lambda y{:}\text{Nat.}\ \text{dbl}\ y) :: Nat \to Nat} \qquad \text{dbl: } x{:}\text{Int} \to \{v{:}\text{Int} \mid v = 2 * x\} \vdash 5 :: Nat$$

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# Another Example

$$Nat \equiv \{v{:}\, Int \mid v > 0\}$$

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$$\frac{\Gamma \vdash e{:}T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e{:}T}$$

$$\frac{\dfrac{}{v = 5 \Rightarrow v > 0}}{\{v{:}\,Int \mid v = 5\} <: \{v{:}\,Int \mid v > 0\}}$$

$$\frac{\text{dbl:}\ x{:}\text{Int} \to \{v{:}\,\text{Int} \mid v = 2 * x\} \vdash 5 :: \{v{:}\,Int \mid v = 5\} \qquad \{v{:}\,Int \mid v = 5\} <: \{v{:}\,Int \mid v > 0\}}{\text{dbl:}\ x{:}\text{Int} \to \{v{:}\,\text{Int} \mid v = 2 * x\} \vdash (\lambda y{:}\text{Nat}.\,\text{dbl}\ y) :: Nat \to Nat \qquad \text{dbl:}\ x{:}\text{Int} \to \{v{:}\,\text{Int} \mid v = 2 * x\} \vdash 5 :: Nat}$$

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# Another Example

$$Nat \equiv \{v{:}\ Int \mid v > 0\}$$

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$$\boxed{\frac{\Gamma \vdash e{:}T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e{:}T}}$$

$\Gamma \equiv$ ┐

$\text{dbl}{:}\ x{:}\text{Int} \to \{v{:}\text{Int} \mid v = 2 * x\}; y{:}Nat \vdash \text{dbl} :: Int \to \{v{:}\text{Int} \mid v = 2 * x\}$ $\qquad \Gamma \vdash Int \to \{v{:}\text{Int} \mid v = 2 * x\} <: Nat \to Nat$

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$$\frac{\Gamma \vdash e\colon T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e\colon T}$$

$\Gamma \equiv$

$$\frac{\Gamma \vdash Nat <: Int \qquad \Gamma; x\colon Nat \vdash \{v\colon \text{Int} \mid v = 2 * x\} <: Nat}{\text{dbl}\colon x\colon \text{Int} \to \{v\colon \text{Int} \mid v = 2 * x\}; y\colon Nat \vdash \text{dbl} :: Int \to \{v\colon \text{Int} \mid v = 2 * x\} \qquad \Gamma \vdash x\colon Int \to \{v\colon \text{Int} \mid v = 2 * x\} <: Nat \to Nat}$$

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dbl: $x{:}\text{Int} \to \{v{:}\text{Int} \mid v = 2 * x\}; y{:}Nat \vdash \text{dbl} :: Int \to \{v{:}\text{Int} \mid v = 2 * x\}$ $\qquad$ $\Gamma \vdash x{:}Int \to \{v{:}\text{Int} \mid v = 2 * x\} <: Nat \to Nat$

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$$y > 0 \wedge x > 0 \wedge v = 2 * x \Rightarrow v > 0$$

$\Gamma \equiv$

$$\Gamma; x{:}\{v{:}Int \mid v > 0\} \vdash \{v{:}\,\text{Int} \mid v = 2 * x\} <: Nat$$

$$\Gamma \vdash Nat <: Int$$

$$\text{dbl}{:}\ x{:}\text{Int} \to \{v{:}\,\text{Int} \mid v = 2 * x\}; y{:}Nat \vdash \text{dbl} :: Int \to \{v{:}\,\text{Int} \mid v = 2 * x\} \qquad \Gamma \vdash x{:}Int \to \{v{:}\,\text{Int} \mid v = 2 * x\} <: Nat \to Nat$$

$$\text{dbl}{:}\ x{:}\text{Int} \to \{v{:}\,\text{Int} \mid v = 2 * x\}; y{:}Nat \vdash \text{dbl} :: Nat \to Nat$$

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$$\text{dbl}{:}\ x{:}\text{Int} \to \{v{:}\,\text{Int} \mid v = 2 * x\} \vdash (\lambda y{:}\,\text{Nat}.\, \text{dbl}\ y) :: Nat \to Nat$$

# Branches

$$\frac{\Gamma \vdash e : bool \quad \Gamma \vdash e_t : \tau \quad \Gamma \vdash e_f : \tau}{\Gamma \vdash if\ e\ then\ e_t\ else\ e_f : \tau}$$

$$\frac{\Gamma \vdash e_1 :: Bool \quad \Gamma; e_1 \vdash e_2 :: T \quad \Gamma; \neg e_1 \vdash e_2 :: T}{\Gamma \vdash if\ e_1\ then\ e_2\ else\ e_3 :: T}$$

Example:

- $\lambda x : Int\ if\ x > 0\ then\ x\ else\ -x\ : Int \to Nat$

# Conventional types are not enough



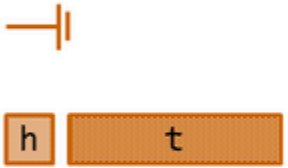
```
// Insert x into a sorted list xs
insert :: x:e  →  xs:List e  →  List e
insert x xs =
  match xs with
    Nil → Nil
    Cons h t →
      if x ≤ h
        then Cons x xs
        else Cons h (insert x t)
```

Code has a bug, but type checker still accepts it.

# Refinement types

```
data SList e where
  Nil :: SList e
  Cons :: h:e →
          t:SList {ν:e | ν ≥ h} →
          SList e
```

sorted lists



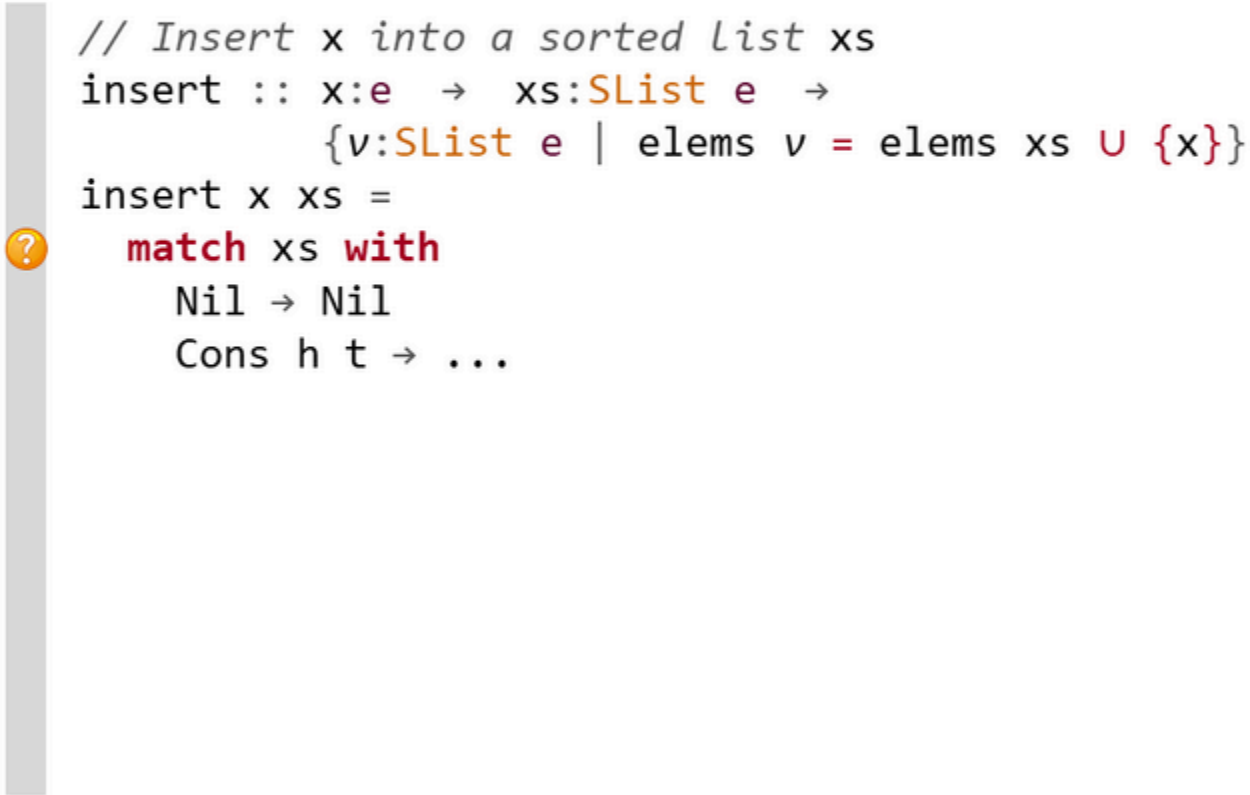
# Refinement types as specs

[Rondon et al. PLDI'08]

```
// Insert x into a sorted list xs
insert :: x:e  →  xs:SList e  →
          {v:SList e | elems v = elems xs ∪ {x}}
insert x xs =
  match xs with
    Nil → Nil
    Cons h t →
      if x ≤ h
        then Cons x xs
        else Cons h (insert x t)
```

Expected
{v:SList e|elems v = elems xs ∪ {x}}
and got
{v:SList e|elems xs ⊆ elems v}

# Incomplete programs?



```
// Insert x into a sorted list xs
insert :: x:e  →  xs:SList e  →
          {v:SList e | elems v = elems xs ∪ {x}}
insert x xs =
  match xs with
    Nil → Nil
    Cons h t → ...
```

# Bidirectional type checking

```
{v:SList e | elems v = {x}}
```



```
insert x xs =
  match xs with
    Nil → Nil
    Cons h t → ...
```

# Round-trip type checking

